TUTORIAL IX

TYPESETTING THEOREMS

IX.1. THEOREMS IN LATEX

In Mathematical documents we often have special statements such as *axioms* (which are nothing but the assumptions made) and *theorems* (which are the conclusions obtained, sometimes known by other names like *propositions* or *lemmas*). These are often typeset in different font to distinguish them from surrounding text and given a name and a number for subsequent reference. Such distinguished statements are now increasingly seen in other subjects also. We use the term *theorem-like statements* for all such statements.

LaTeX provides the declaration `\newtheorem` to define the theorem-like statements needed in a document. This command has two arguments, the first for *the name we assign to the environment* and the second, *the name to be printed with the statement*. Thus if you want

Theorem 1. *The sum of the angles of a triangle is* 180°.

you first specify

```
\newtheorem{thm}{Theorem}
```

and then type

```
\begin{thm}
  The sum of the angles of a triangle is $180^\circ$.
\end{thm}
```

Note that in the command `\newtheorem` the first argument can be any name you fancy, instead of the `thm` given here. Also, it is a good idea to keep all your `\newtheorem` commands together in the preamble.

The `\newtheorem` command has a couple of optional arguments which control the way the corresponding statement is numbered. For example if you want the above theorem to be numbered 1.1 (the first theorem of the first section) rather than a plain 1, then you must specify

```
\newtheorem{thm}{Theorem}[section]
```

in the `\newtheorem` command. Then the same input as above for the theorem produces

Theorem IX.1.1. *The sum of the angles of a triangle is* 180°.

The next **Theorem** will be numbered 1.2, the third **Theorem** in the fourth section will be numbered 4.3 and so on.

The other optional argument of the `\newtheorem` command is useful when you have several different types of theorem-like statements (such as lemmas and corollaries) and you want some of them to share the same numbering sequence. For example if you want

Theorem IX.1.2. *The sum of the angles of a triangle is* 180°.

An immediate consequence of the result is the following

Corollary IX.1.3. *The sum of the angles of a quadrilateral is* 360°.

Then you must specify

```
\newtheorem{cor}[thm]{Corollary}
```

after the specification `\newtheorem{thm}[section]` and then type

```
\begin{thm}
  The sum of the angles of a triangle is $180^\circ$.
\end{thm}
```

An immediate consequence of the result is the following

Corollary IX.1.4. *The sum of the angles of a quadrilateral is* 360°.

The optional argument `thm` in the definition of the `cor` environment specifies that "Corollaries" and "Theorems" are to be numbered in the same sequence.

A theorem-like environment defined using the `\newtheorem` command has also an optional argument which is used to give a *note* about the theorem such as the name of its discoverer or its own common name. For example, to get

Theorem IX.1.5 (Euclid). *The sum of the angles of a triangle is* 180°.

you must type

```
\begin{thm}[Euclid]
  The sum of the angles of a triangle is $180^\circ$.
\end{thm}
```

Note the optional argument `Euclid` after the `\begin{thm}`. This use of [...] for optional notes sometimes lead to unintended results. For example, to get

Theorem IX.1.6. $[0,1]$ *is a compact subset of* $\mathbb{R}$.

if you type

```
\begin{thm}
  [0,1] is a compact subset of $\mathbb{R}$.
\end{thm}
```

then you get

Theorem IX.1.7 (0,1). *is a compact subset of* $\mathbb{R}$.

Do you see what happened? The string `0,1` within [ ] at the beginning of the theorem is considered an optional note by LaTeX ! The correct way is to type